

UCE312 :SOLID MECHANICS

L	T	P	Cr
3	1	2*	4.0

*** Lab every alternate week**

Course Objective: This subject introduces fundamental concepts of structural mechanics to solve general engineering mechanics problems. It aims to develop an understanding of the stresses strains and deformations that develop in solid materials when they are subjected to different types of loading and to develop an understanding of the conditions at failure of such materials.

Introduction of equilibrium of bodies: Concept of Free-body diagrams and conditions of equilibrium of bodies, static determinacy.

Plane Trusses: Forces in members of a truss by method of joints and method of sections.

Axial Stress and Strain: Concept of stress, strain, elasticity and plasticity; one-dimensional stress-strain relationships; statically determinate and indeterminate problems, compound and composite bars.

Shear Force and Bending Moment Diagrams: Types of load on beams, classification of beams; axial, shear force and bending moment diagrams: simply supported, overhanging and cantilever beams subjected to any combination of point loads, uniformly distributed and varying load and moment.

Bending & Shear Stresses in beams: Derivation of flexural formula for straight beams, concept of centroid and second moment of area, bending stress calculation for beams of simple and built up sections, flitched beams. Shear stress formula for beams, shear stress distribution in beams.

Deformations: Governing differential equation for deflection of straight beams having constant flexural rigidity, Double Integration, Macaulay's, Moment Area and Conjugate Beam method for determining slopes and deflection in beams.

Laboratory Work

1. To find out the tensile strength of ductile materials (MS/Al) and plot its stress-strain characteristics.
2. To determine impact strength of ductile and brittle materials.
3. Experimental verification of theory of bending (calculation of bending stress and deflections at various points in the beam theoretically and verifying the same experimentally) and indirect evaluation of the modulus of elasticity.
4. To determine the hardness of different types of materials

Course Learning Objectives (CLO)

Upon completion of this course, the students will be able to:

1. Identify and quantify all forces associated with a static framework/ plane force systems.
2. Evaluate axial stresses and strains in various determinate and indeterminate structural systems.
3. Draw shear force diagram and bending moment diagram in various kinds of beams subjected to different kinds of loads.
4. Evaluate the different types of stresses (Bending/Shear) in flexural members.
5. Determine deformations and deflections in various kinds of beams.

Text Books

1. Engineering Mechanics of Solids, E.P. Popov and T.A. Balan, Pearson India Education Services, 2018.
2. Strength of Materials, D.K. Singh, Springer International Publishing, Fourth edition, 2020.

Reference Books

1. Mechanics of materials, R. C. Hibbeler, Pearson, Tenth edition, 2016.
2. Mechanics of materials, F.P. Beer, E.R. Johnston, D. Mazurek, McGraw-Hill Higher Education, 2011.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessment and Quizzes, Project (Including report, presentation etc.)	40